

Projected Yukawa Line and Generation-Level Assignment

Functorial Closure of $L_Y = \wedge^2(\mathcal{S}_\Pi)$ and Spectral Ordering on $\mathbb{C}_{\text{gen}}^3$

J. Beau

Independent Researcher, France

`jerome.beau@cosmochrony.org`

2026

Abstract

This note opens the mass-sector frontier of the fermionic-matter sub-programme of Cosmochrony, and closes only its first stage. It does not derive the three-generation masses, the Yukawa couplings, or the mixing. It isolates two structural results that the admissible spinor carrier already determines, and separates them sharply from the mass normalisation that does not yet follow. First, on the rank-two admissible spinor bundle $\mathcal{S}_\Pi$ produced by the Lorentzian complexification of the metaplectic Weil carrier [4], the determinant line $L_Y := \wedge^2(\mathcal{S}_\Pi)$ is the unique functorial line of $\mathcal{S}_\Pi$ that is invisible to the adjoint sector $\text{Sym}^2(\mathcal{S}_\Pi) \simeq \text{ad}_{\mathbb{C}}(P_\Pi)$, and it is the carrier of the abelian electroweak weight; the projected Yukawa sector is therefore the determinant-line sector of the admissible spinor functorial closure, not an external bundle datum. Second, on the gauge-singlet generation triplet $\mathbb{C}_{\text{gen}}^3 = \text{span}(e_0, e_+, e_-)$ the squared projective residue takes the closed normal form $E_\Pi^2|_{\mathbb{C}_{\text{gen}}^3} = \text{diag}(1, \frac{1}{2} + u, \frac{1}{2} - u)$ [7], whose exit deficits $\{0, \frac{1}{2} - u, \frac{1}{2} + u\}$ order the three generation levels [4]. The single split coefficient u fixes the dimensionless level structure but not the absolute scale: the projected Yukawa operator, its norm, the sign of u , and the complex mixing phase remain downstream open data, so no mass value is claimed. All representation-theoretic and spectral identities are verified by exact symbolic computation, with no sampling.

Keywords: projected Yukawa sector; determinant line; admissible spinor bundle; tensor functors; hypercharge line; generation ordering; exit deficit;

squared projective residue; spectral level assignment; non-injective projection.

Contents

1	Position and purpose	4
2	The determinant line L_Y	4
3	Separation from the Yukawa operator	5
4	Generation-level assignment	6
5	Status of results	7
6	Reproducibility	7

1 Position and purpose

The fermionic-matter sub-programme identifies the electroweak and generation structure of matter from the admissible spinor bundle $\mathcal{S}_\Pi$ alone, without an external matter bundle. Three identifications are already established in the constituent paper [4]: the tensor functors of $\mathcal{S}_\Pi$ reproduce the $SU(2)_L \times U(1)_Y$ electroweak bundle (Theorem A), the projected Dirac residue E_Π enforces the $V - A$ chirality and fixes hypercharge as a spectral coherence datum (Theorem B), and the saturation invariant forces a gauge-singlet three-generation factor $\mathbb{C}_{\text{gen}}^3$ (Theorem C). The split coefficient of $\mathbb{C}_{\text{gen}}^3$ has since been reduced to a single real number u , with its existence $u \neq 0$ closed in the projective-residue Schur reduction [7] and the radial factor of its modulus closed in the Born–Infeld margin note [6]. The sub-programme note states plainly that the mass spectrum and the Yukawa couplings are not yet derived, and lists them among the open deliverables [1].

This note takes up the first stage of that open deliverable, and only the first stage. It does not compute a mass. It closes two structural statements that the carrier already determines — the functorial Yukawa line and the generation-level ordering — and it isolates, as the genuinely open downstream object, the projected Yukawa operator whose norm and phase carry the absolute mass scale and the mixing. The method lock is stated once and held throughout: the determinant line L_Y fixes the coupling line, the squared residue $E_\Pi^2|_{\mathbb{C}_{\text{gen}}^3}$ fixes the generation levels, and the mass comes after.

2 The determinant line L_Y

The carrier is the rank-two admissible spinor bundle $\mathcal{S}_\Pi$, the fundamental bundle of $\text{Spin}(3, 1) \simeq \text{SL}(2, \mathbb{C})$ obtained from the metaplectic lift of the Weil module $\mathcal{V}_{\mathbb{R}}$ under the Lorentzian complexification $\text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$ forced by $g^{\mu\nu} = 2\eta^{\mu\nu}$ [4]. The tensor functors of $\mathcal{S}_\Pi$ are taken on the rank-two fibre, not on the doubled Dirac space $\mathcal{S}_L \oplus \mathcal{S}_R$, so that no cross-chiral components are manufactured.

Proposition 1 (Projected Yukawa line). *Let $\mathcal{S}_\Pi$ be the rank-two admissible spinor bundle obtained from the Lorentzian complexification of the metaplectic Weil carrier. Then the determinant line*

$$L_Y := \wedge^2(\mathcal{S}_\Pi)$$

is the unique functorial line produced by $\mathcal{S}_\Pi$ that is invisible to the $SU(2)_L$ adjoint sector $\text{Sym}^2(\mathcal{S}_\Pi) \simeq \text{ad}_{\mathbb{C}}(P_\Pi)$, and it carries the abelian electroweak

weight. Consequently the projected Yukawa sector is not an additional external bundle datum, but the determinant-line sector of the admissible spinor functorial closure.

Proof. On the rank-two fibre $\mathbb{C}^2$ the polynomial functors are the symmetric and exterior powers Sym^k and $\wedge^k$. Their dimensions are $\dim \text{Sym}^2(\mathbb{C}^2) = 3$ and $\dim \wedge^2(\mathbb{C}^2) = 1$, and $\wedge^2(\mathbb{C}^2)$ is the only nontrivial one-dimensional instance, so $L_Y = \wedge^2(\mathcal{S}_\Pi)$ is the unique functorial line of $\mathcal{S}_\Pi$. The symmetric sector matches the complex adjoint, $\text{Sym}^2(\mathbb{C}^2) \simeq \mathfrak{sl}_2(\mathbb{C})$ of dimension 3, whose compact real form is the $\text{SU}(2)_L$ gauge sector [4]. For any $g \in \text{SL}(2, \mathbb{C})$, and in particular for the compact real form $\text{SU}(2)_L \subset \text{SL}(2, \mathbb{C})$ selected from $\text{Sym}^2(\mathcal{S}_\Pi)$, the induced action on the line $\wedge^2(\mathbb{C}^2)$ is multiplication by $\det(g) = 1$. Hence L_Y carries no $\text{SU}(2)_L$ charge: it is invisible to the adjoint sector. The induced action of a diagonal abelian generator $\text{diag}(y_1, y_2)$ on $\wedge^2(\mathbb{C}^2)$ is multiplication by the trace $y_1 + y_2$, so the determinant line is the carrier of the abelian weight, with $\text{U}(1)_Y$ realised as the structure group $\text{GL}(1)$ of the line. The identification $\wedge^2(\mathcal{S}_\Pi) = L_Y$ is Theorem A of [4], and no electroweak matter bundle is introduced beyond the functors of $\mathcal{S}_\Pi$. $\square$

The determinant line is thus intrinsic to the carrier. It is the abelian half of the same functorial closure whose symmetric half is the non-abelian gauge sector, and it requires no separate hypercharge bundle.

3 Separation from the Yukawa operator

Proposition 1 fixes a line, not a mass matrix. The line L_Y is the weight datum of the Yukawa coupling; the mass matrix is a distinct object, a projected morphism of the admissible spinor sectors of the schematic form

$$Y_\Pi : \mathcal{S}_{L,\Pi} \otimes L_Y \longrightarrow \mathcal{S}_{R,\Pi}, \quad (1)$$

or equivalently its spectral avatar after compression by the projective residue E_Π . The line fixes which abelian weight the coupling carries; the operator Y_Π fixes how strongly, and in which generation directions, the chiralities are tied. This separation is the central guardrail of the note: the closure of the coupling line does not close the operator. The norm of Y_Π , the orientation (sign) of the split coefficient u , and the complex metaplectic mixing phase are downstream data, recorded as open in the sub-programme inventory [1]; none of them is supplied here.

4 Generation-level assignment

The internal structure of the gauge-singlet triplet $\mathbb{C}_{\text{gen}}^3$ is read from the squared projective residue E_{Π}^2 , the zero-order spectral residue of non-injective projection in the spinorial sector [4, 7]. On $\mathbb{C}_{\text{gen}}^3 = \text{span}(e_0, e_+, e_-)$ with Cartan generator $J_3 = \text{diag}(0, 1, -1)$, the Schur reduction fixes the normal form $E_{\Pi}^2|_{\mathbb{C}_{\text{gen}}^3} = (C_2 - J_3^2)/C_2 + u J_3 + v R_{\text{mix}}$ with spin-one Casimir $C_2 = 2$ [7]. The CP-even real sector imposes $v = 0$, and the even part is closed analytically by Born–Infeld parity on the equatorial admissible fibre, $(C_2 - J_3^2)/C_2 = \text{diag}(1, \frac{1}{2}, \frac{1}{2})$ [3].

Proposition 2 (Generation-level assignment). *On the gauge-singlet generation triplet $\mathbb{C}_{\text{gen}}^3 = \text{span}(e_0, e_+, e_-)$, the projected Yukawa levels are ordered by the spectral levels of*

$$E_{\Pi}^2|_{\mathbb{C}_{\text{gen}}^3} = \text{diag}\left(1, \frac{1}{2} + u, \frac{1}{2} - u\right).$$

With the central J_{Π} -fixed level $s_0 = 1$, the exit deficits $d_{\text{exit}}(e_i) := s_0 - s_i$ are

$$d_{\text{exit}}(e_0) = 0, \quad d_{\text{exit}}(e_+) = \frac{1}{2} - u, \quad d_{\text{exit}}(e_-) = \frac{1}{2} + u,$$

so for $u > 0$ the ordering $0 < d_{\text{exit}}(e_+) < d_{\text{exit}}(e_-)$ assigns e_0 the lightest and e_- the heaviest generation level. The assignment determines the three generation levels structurally. It does not by itself determine the physical mass normalisation, which still depends on the projected Yukawa norm and the upstream dictionary data.

Proof. The normal form is [7] with $v = 0$ and the even closure [3]; its eigenvalues on (e_0, e_+, e_-) are $(1, \frac{1}{2} + u, \frac{1}{2} - u)$. The exit deficits follow from the support-exit definition relative to the non-exiting central level $s_0 = 1$ [4, 2]. The ordering for $u > 0$ follows from $d_{\text{exit}}(e_-) - d_{\text{exit}}(e_+) = 2u > 0$, which places the central direction e_0 at zero deficit (lightest, non-exiting) and e_- at the largest deficit (heaviest). $\square$

The level structure is dimensionless and fixed by u alone: the ratio $(\frac{1}{2} + u)/(\frac{1}{2} - u)$ is invariant under the spectral rescaling $E_{\Pi}^2 \mapsto \lambda E_{\Pi}^2$ [7], so the generation ordering does not depend on the absolute spectral normalisation. The numerical value of u is itself dictionary-bound through the chiral-frontier normalisation $\mathcal{N}_A$ [5], and is not fixed in this note; what is fixed here is the assignment map $e_0, e_+, e_- \mapsto$ three ordered generation levels, prior to any identification with the physical e, μ, τ .

5 Status of results

Derived (this note). The functorial Yukawa line $L_Y = \wedge^2(\mathcal{S}_\Pi)$ is the unique line of the carrier, invisible to the $SU(2)_L$ adjoint sector and carrying the abelian weight (Proposition 1). The generation-level assignment on $\mathbb{C}_{\text{gen}}^3$ is fixed by the spectral levels of $E_\Pi^2|_{\mathbb{C}_{\text{gen}}^3} = \text{diag}(1, \frac{1}{2} + u, \frac{1}{2} - u)$ through the exit deficits (Proposition 2). Both are structural and unconditional given the carrier identifications of [4, 7, 3].

Open. The projected Yukawa operator Y_Π of (1), its norm, the orientation (sign) of u , and the complex metaplectic mixing phase are not supplied here. Hence the absolute masses, the mass ratios beyond the dimensionless level structure, and the generation mixing remain open, in agreement with the open deliverables of [1].

Summary. Front 3a is closed as a structural result: the Yukawa line and the generation ordering, not the mass values. The determinant line L_Y fixes the coupling line; the squared residue $E_\Pi^2|_{\mathbb{C}_{\text{gen}}^3}$ fixes the levels; the mass comes after.

6 Reproducibility

All representation-theoretic and spectral identities are verified by exact symbolic computation, with no floating-point sampling, in `simulation/fermionic-matter/front3_yukawa`. The functorial dimensions $\dim \text{Sym}^2(\mathbb{C}^2) = 3$ and $\dim \wedge^2(\mathbb{C}^2) = 1$, the $SL(2, \mathbb{C})$ action on the line $\wedge^2(\mathbb{C}^2)$ as $\det = 1$, and the abelian weight as the trace $y_1 + y_2$ are checked symbolically (Proposition 1). The normal form $E_\Pi^2|_{\mathbb{C}_{\text{gen}}^3} = \text{diag}(1, \frac{1}{2} + u, \frac{1}{2} - u)$, its even-sector reduction to $\text{diag}(1, \frac{1}{2}, \frac{1}{2})$ at $u = 0$, the exit deficits, the ordering for $u > 0$, and the scale invariance of the level ratio are checked symbolically (Proposition 2).

References

- [1] J. Beau. The Fermionic Matter Sub-Programme, 2026. Presentation Note 6: derives fermions, chirality, hypercharge, and three generations as the spinorial face of the admissible Weil module V_ρ after Lorentzian complexification of its metaplectic lift. Three structural results (Theorems A–C of Q14): $\text{Sym}^2(\mathcal{S}_\Pi) \simeq \text{ad}_\mathbb{C}(P_\Pi)$ and $\wedge^2(\mathcal{S}_\Pi) = L_Y$ reproduce $\text{SU}(2)_L \times \text{U}(1)_Y$; the projective endomorphism E_Π enforces $V - A$ chirality and hypercharge rigidity (cond. on Hypothesis 3.5); $\sigma_c(n_3) = 3$ has a spinorial reading giving a gauge-singlet three-generation factor C_{gen}^3 , doi:10.5281/zenodo.20562665.
- [2] J. Beau. Three Particle Generations and Mass Hierarchy from Spectral Stratigraphy, 2026. Spectral admissibility sub-program paper, doi:10.5281/zenodo.19040490.
- [3] Jérôme Beau. Born–Infeld Parity Forces an Equatorial Admissible Fibre: Analytical Derivation of the $[1 : \frac{1}{2} : \frac{1}{2}]$ Eigenvalue Ratio in $\text{Sym}^2(V_\rho)$, 2026. "O30" paper, doi:10.5281/zenodo.20014707.
- [4] Jérôme Beau. Fermionic Matter and Chirality from Projective Dirac Admissibility, 2026. "Q14" paper, doi:10.5281/zenodo.20218409.
- [5] Jérôme Beau. Reduction of the Angular Amplitude Observable: the J_3 Split as Accumulated Oriented Symplectic Area, 2026. Fermionic matter sub-programme working note: identifies, before any Weil–BFS computation, the numerical observable for the absolute normalisation of the angular generation split of Q14 §6. A Baker–Campbell–Hausdorff reduction identifies the per-step J_Π -odd J_3 component of the metaplectic cascade generator with the oriented symplectic area swept per step (Carnot-degree-2 central increment of Heis_3), not a free shear parameter. The quantity to be measured is the accumulated oriented J_3 -odd area along the Weil–BFS cascade, normalised by the projected capacity $\hat{I}(n)$; at n_3^{obs} this equals $\varepsilon_{\text{Weil}}(n_3^{\text{obs}})$, to be compared with the dictionary value $\varepsilon = 1/10$ without fitting. In the O12 fingerprint the two sectors are the two terms of the exponent: frequency B_c carries the radial capacity σ_c , central phase A_c carries the angular split. Direct evaluation on symmetric BFS shells returns $\Theta_{\text{raw}} = 0$ identically; the orientation prescription is an irreducible input prior to $\mathcal{N}_A$, doi:10.5281/zenodo.20601228.
- [6] Jérôme Beau. The Born–Infeld Saturation Margin of the Chiral Modulus: Antisymmetric Structure and the Reduction of the Generation Split to

a Lorentzian Genus, 2026. Fermionic matter sub-programme companion note to PRS. Defines the Lorentzian saturation functional $\mathcal{B}_{\text{sat}}(s)$ along the J_{Π} -odd modulus controlling the split coefficient u , and reduces its second variation to $\mu_{\chi}^2 := \partial_s^2 \mathcal{B}_{\text{sat}}(0) = \frac{1}{2} P_{\chi, \mu\nu} P_{\chi}^{\mu\nu}$; in the Schur-transverse branch closed in PRS the chiral polarisation is electric, so $\mu_{\chi}^2 < 0$ and the split opens spontaneously, doi:10.5281/zenodo.20633931.

- [7] Jérôme Beau. The Projective Residue as a Schur Complement: Reduction of the Generation Split to the Chiral Asymmetry of Projection Locking, 2026. Fermionic matter sub-programme note: exhibits the projective endomorphism as the Schur complement $E_{\Pi} = -\Pi_S D (1 - P) D \Pi_S^* = -M^{\dagger} M$ of the spinorial directions eliminated by the non-injective projection; reduces the inter-generation split coefficient u to the $\mathcal{D}^{\pm}$ -transported, generation-projected component of the antiunitary chiral equivariance defect $\Delta_{\chi}(P) = \pi_{LL} - \tau \bar{\pi}_{R\bar{R}} \tau^{-1}$; stratifies the result on the Cosmochrony axioms (A1–A3 force $u = 0$; non-zero u requires A4 chiral symmetry-breaking); proves finite/Lorentzian separation (finite locking is J_{Π} -equivariant, so $u_{\text{fin}} = 0$); locks Seeley–DeWitt conventions so operator and spectral u coincide; no-go result: chirality is Lorentzian, not a finite-fibre label. Reduces the open problem to a Lorentzian computation of the chiral curvature $\mu_{\chi}^2 := \partial_s^2 \mathcal{B}(0)$ along the J_{Π} -odd modulus of the saturated Born–Infeld functional, doi:10.5281/zenodo.20601040.